Learning Signed Critic Errors: A Probabilistic Approach to Addressing Function Approximation Errors in Actor–Critic Reinforcement Learning

Blue Sky Ideas Track

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Abstract

Actor-critic reinforcement learning relies on learned value estimates to guide policy optimization. However, function approximation errors can propagate through temporal-difference learning, producing inaccurate critic estimates and potentially suboptimal policy updates. Methods such as Twin Delayed Deep Deterministic Policy Gradient (TD3) address these effects through clipped double critics, delayed policy updates, and target-policy smoothing, but do not explicitly learn state–action dependent corrections from observed return discrepancies. This creates a fundamental problem of determining how critic estimation errors can be identified and corrected to support more reliable policy improvement. We investigate a probabilistic approach in which Gaussian-process regression (GPR) is used to model the signed discrepancies between neural critic predictions and Monte Carlo rollout estimates. The posterior mean estimates the residual correction, while predictive variance controls its magnitude, allowing adjustments to both overestimated and underestimated values. We examine this procedure with Deep Deterministic Policy Gradient (DDPG), TD3, and Soft Actor-Critic (SAC) on Pendulum-v1. Evaluations across five seeds show reductions in average checkpoint-level discrepancies for all three algorithms, although these measurements are restricted to the samples used for regression fitting and do not establish improved control performance. These findings motivate the development of actor-critic methods that use independently evaluated residual estimates for value learning and policy updates. Such methods could provide a foundation for more dependable continuous-control learning in autonomous robotic systems, where errors in critic predictions can influence subsequent control decisions.

CCS Concepts

• **Computing methodologies → Reinforcement learning**.

Keywords

Actor-critic reinforcement learning, signed critic errors, Gaussian process regression, probabilistic correction



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1 Introduction and Background

Actor-critic reinforcement learning uses a learned value function to guide policy improvement [21]. The actor selects actions, while the critic estimates their long-term returns [20]. When the critic is inaccurate, the actor may prefer actions whose estimated values are not supported by their outcomes. This is more critical in continuous control because the actor learns from the critic's action gradients [18].

Preliminary Information and Related Literature: In actor-critic reinforcement learning, policy improvement depends on the accuracy of a learned action-value function [12]. For a policy π, the expected discounted return is

$$Q^{\pi}(s,a) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid s_0 = s, a_0 = a\right], \tag{1}$$

where $\gamma \in [0,1)$ is the discount factor. Since exact action values are generally unavailable, deep reinforcement learning approximates them using neural critics trained through temporal-difference (TD) learning [20]. A typical critic minimizes the following loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{D}}\left[(Q_\theta(s,a) - y)^2\right], \tag{2}$$

where y is a bootstrapped target that itself depends on estimated future values. Different actor-critic algorithms construct this target differently. DDPG [11] uses a single target critic:

$$y_{\text{DDPG}} = r + \gamma(1-d)Q_{\bar{\theta}}(s', \mu_{\bar{\phi}}(s')). \tag{3}$$

TD3 [5] reduces overestimation through clipped double critics and target-policy smoothing:

$$y_{\text{TD3}} = r + \gamma(1-d)\min_{j=1,2} Q_{\bar{\theta}_j}(s', \tilde{a}'), \tag{4}$$

where $\tilde{a}'$ is the noise-perturbed target action. SAC [6] uses an entropy-regularized target:

$$y_{\text{SAC}} = r + \gamma(1-d)\Big[\min_{j=1,2} Q_{\bar{\theta}_j}(s', a') - \alpha \log \pi_\phi(a'|s')\Big], \tag{5}$$

where $a' \sim \pi_\phi(\cdot|s')$ and α controls entropy regularization.

Despite these differences, all three methods bootstrap from learned critics. If an estimated action value differs from its corresponding true objective by $\epsilon(s, a)$, this error can propagate through subsequent TD targets [27]. A single critic, as in DDPG, can propagate both overestimation and underestimation. Taking the minimum of two critics, as in TD3 and SAC, reduces susceptibility to overestimation but can also introduce underestimation [13, 22]. For SAC, the relevant objective additionally includes entropy, making reward-only returns an imperfect reference for its soft critic [7]. Several methods address value estimation bias [24, 28]. Double Q-learning [25] uses separate estimators for action selection and evaluation, while Double Deep Q-Network (Double DQN) [26] uses online and target networks for these operations. Truncated Quantile Critics (TQC) [10] removes upper quantiles from target distributions to reduce overestimation. Randomized Ensembled Double Q-Learning (REDQ) [3] computes target values using the minimum estimate from a random subset of critics. CrossQ [2] improves critic updates through batch normalization without target networks.

Although existing methods improve value estimation, they do not explicitly model signed, state-action dependent discrepancies between critic predictions and observed returns. More broadly, autonomous agents rely on learned value estimates to make decisions, where both overestimation and underestimation can influence individual behaviour and, in multiagent settings, collective outcomes. Our vision moves beyond conventional bias-reduction strategies by treating signed estimation errors as a distinct probabilistic learning signal. This opens a broader research direction toward error-aware agents that learn not only the value of their actions but also when and how to correct their estimates, potentially supporting more reliable decision-making, adaptation, and coordination in dynamic multiagent environments.

2 Blue Sky Vision and Real-World Impact

Blue Sky Vision: Just like a high critic estimate, a low critic estimate could be wrong. The question is how an agent can learn the direction and magnitude of its own value estimation errors and determine when a correction is reliable [9]. Rather than only changing how value estimates are computed, our vision is to develop actor–critic agents that learn both action values and their own estimation errors. We explore this idea using Gaussian process regression (GPR), where the posterior mean estimates signed discrepancies and predictive variance controls the correction strength [17]. The broader idea is that an agent could identify where its critic is overestimating or underestimating and use independently validated corrections during learning. This would allow future actor-critic methods to learn not only from estimated returns but also from the errors in those estimates. The long-term vision is to move beyond agents that merely learn action values toward agents that explicitly learn from their own estimation errors, treating these errors as a distinct learning signal for deciding when to trust, correct, or reconsider their value estimates. This opens a broader research direction toward self-evaluating agents capable of adapting their learning and decision-making in uncertain and changing environments.

Real-World Impact: This direction could be useful in robotic manipulation, autonomous navigation, and other continuous-control applications. For example, a robotic arm may overestimate the value of a grasp that is likely to fail or underestimate a grasp that could succeed. Similarly, a navigation agent may prefer a poor route because its critic assigns an excessively high value to certain actions. Learning signed critic errors could help such agents recognize these discrepancies and adjust their value estimates before using them for policy improvement. The applications extends to multiagent systems, where several autonomous agents interact and make decisions in a shared environment. For example, in multi-robot navigation, one robot may overestimate the value of moving through a path without considering how the actions of other robots may affect its outcome. In collaborative robotic tasks, an agent may underestimate the value of waiting or cooperating with another agent. Since the behaviour of other agents can change during learning, previously learned value estimates may also become inaccurate [1]. Learning signed critic errors could help each agent identify such discrepancies and adapt its value estimates as interactions change.

3 Probabilistic Modeling and Correction of Critic Errors

The main idea of this work comes from a simple observation: in actor-critic reinforcement learning, the critic may overestimate or underestimate the value of an action, and both can affect learning [5, 20]. In this approach, we use Gaussian Process Regression (GPR) [17] to model the discrepancies between neural critic predictions and Monte Carlo rollout returns. Along with the direction and magnitude of these errors, we consider the uncertainty in the regression predictions so that uncertain corrections have less influence. We study this approach using Deep Deterministic Policy Gradient (DDPG), Twin Delayed Deep Deterministic Policy Gradient (TD3), and Soft Actor-Critic (SAC) in the Pendulum-v1 environment [4].

3.1 Learning Signed Critic Discrepancies

DDPG, TD3, and SAC learn from observed transitions $< s, a, r, s', d >$ stored in a replay buffer, where s and a denote the state and action, r is the reward, s' is the next state, and d indicates episode termination. The algorithms differ in how they compute target Q-values during critic updates, as shown in Eqs. (3), (4), and (5). DDPG uses a single target critic, while TD3 uses the minimum of two target critics with target-policy smoothing to reduce overestimation, although this may introduce underestimation. SAC also uses two critics but includes an entropy term in its target value. We investigate Gaussian process regression (GPR) as an auxiliary probabilistic model for estimating discrepancies between neural critic predictions and Monte Carlo rollout references. At each evaluation checkpoint, policy is fixed, and $n = 50$ observations are sampled from the replay buffer. For each observation s_i, the deterministic policy supplies an action a_i. The critic estimate is

$$q_i = \begin{cases} Q_{\theta_1}(s_i, a_i), & \text{DDPG}, \\ \min\limits_{j=1,2} Q_{\theta_j}(s_i, a_i), & \text{TD3 or SAC}. \end{cases} \tag{6}$$

The Monte Carlo reference is obtained by averaging discounted rollout returns [20]:

$$\widehat{G}_i = \frac{1}{M} \sum_{m=1}^{M} \sum_{t=0}^{H_m - 1} \gamma^t r_{i,m,t}, \tag{7}$$

where $M = 10$ and $\gamma = 0.99$ in the recorded Pendulum-v1 environment evaluations. The signed discrepancy between the critic estimate and the rollout reference is defined as

$$e_i = q_i - \widehat{G}_i. \tag{8}$$

A positive discrepancy indicates that the critic estimate exceeds the reference, whereas a negative discrepancy indicates that it falls below the reference. Preserving this sign is important because while correcting an overestimated value we need to decreases it and while correcting an underestimated value we need to increases it.

3.2 Probabilistic Correction Using GPR

Once the signed discrepancies are obtained, we use GPR to learn their relationship with the corresponding state–action pairs. We use a Radial Basis Function (RBF) kernel [17]:

$$k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right), \tag{9}$$

where $x = (s, a)$ represents a state–action pair and ℓ denotes the length-scale parameter. After fitting the model, GPR provides a posterior mean $\mu_e(x)$ and a predictive standard deviation $\sigma_e(x)$ [17]. The mean estimates the direction and magnitude of the signed discrepancy defined in Eq. (8), while the standard deviation indicates the uncertainty in this estimate.

The corrected Q-value is calculated as:

$$Q_c(x) = q(x) - \lambda(x)\mu_e(x), \tag{10}$$

where $q(x)$ is the critic estimate defined in Eq. (6), and $\mu_e(x) \in \mathbb{R}$ is the GPR posterior mean of the signed discrepancy. The correction strength is controlled by

$$\lambda(x) = \exp\left(-\frac{\sigma_e^2(x)}{2\tau^2}\right), \qquad \tau = 5, \tag{11}$$

where $\lambda(x) \in (0, 1]$ reduces the correction when predictive uncertainty is high.

When $\mu_e(x) > 0$, the Q-value decreases, while for $\mu_e(x) < 0$, the Q-value increases. The corrected discrepancy is computed as:

$$e_c(x) = Q_c(x) - \widehat{G}(x), \tag{12}$$

where $\widehat{G}(x)$ is the Monte Carlo rollout return estimate defined in Eq. (7).

3.3 Proposed GPR-Guided Actor–Critic Learning

The current work applies GPR without changing the old mechanism of DDPG, TD3 and SAC. In Actor-Critic Algorithms critic updates depends on target values and when these targets contain errors then those error may cause big impact [5, 20], so, we are using an independently validated GPR model to estimate the signed discrepancy. For an actor-critic algorithm $\mathcal{A}$, the proposed corrected target is:

$$y_{\mathcal{A}}^{\text{GPR}} = y_{\mathcal{A}} - \gamma(1 - d)\lambda^-(x')\mu_e^-(x'). \tag{13}$$

where $x' = [s', a']$ denotes the next state–action pair, with s' obtained from the replay buffer and a' generated by the corresponding policy. Since the TD target depends on the estimated value of the next state–action pair, errors in this estimate can propagate during critic updates. Therefore, we propose using the GPR-predicted signed discrepancy $\mu_e^-(x')$ to adjust the target value, while $\lambda^-(x')$ reduces the correction when uncertainty is high. For DDPG, the residual model would estimate discrepancies associated with its single target critic. In TD3, it would be aligned with the minimum of two target critics, while in SAC, it would also need to account for the entropy-regularized value objective. The goal is to develop actor–critic methods that learn not only action values but also how to correct their own value estimates.

Table 1: Absolute checkpoint-mean discrepancy across five seeds.

	Raw	GPR	Reduction
DDPG	3.212 ± 0.648	2.007 ± 0.586	37.5%
TD3	1.240 ± 0.357	0.486 ± 0.230	60.8%
SAC	2.114 ± 0.686	1.116 ± 0.990	47.2%

4 Preliminary Findings Towards Generalized Critic Error Correction

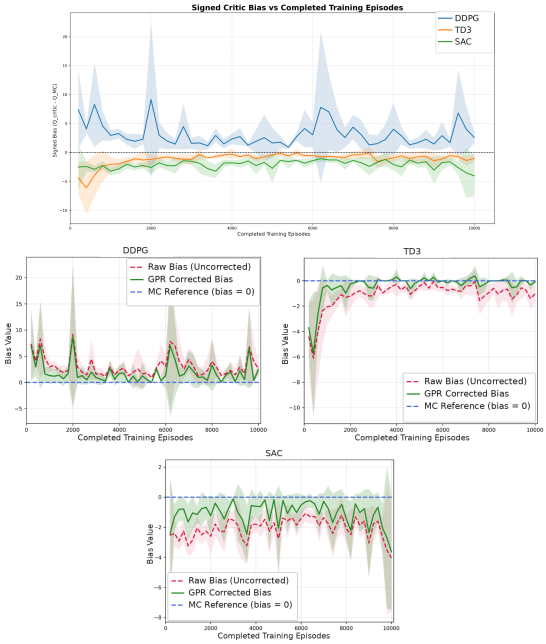


Figure 1: Signed critic–reference discrepancies on Pendulum-v1. Uncorrected bias and GPR-corrected discrepancies for DDPG, TD3, and SAC. Lines show five-seed means and shaded regions indicate variability. Corrections are evaluated on fitting samples.

Protocol and metric: The Pendulum-v1 records contain 750 evaluations: three algorithms, five seeds (0–4), and 50 checkpoints from episode 200 to 10,000. Each run contains two million training transitions. Training uses Stable-Baselines3 MLP policies [16]. DDPG

and TD3 use Gaussian exploration noise with standard deviation 0.1. TD3 uses policy delay 2, target noise 0.2 and noise clipping 0.5; SAC learns its entropy coefficient. Other settings use the installed defaults. Package versions were not recorded, limiting reproducibility [8, 14]. Each logged discrepancy is already averaged across sampled states. For algorithm a and seed s, we calculate:

$$A_{a,s} = \frac{1}{50} \sum_{k=1}^{50} |b_{a,s,k}|, \tag{14}$$

where $b_{a,s,k}$ is the signed checkpoint mean. Table 1 reports the mean and sample standard deviation of $A_{a,s}$ across seeds. Positive and negative errors may cancel when averaged at each checkpoint. Therefore, the reported values do not represent the absolute error of individual state-action pairs.

Observations and Analysis: The mean absolute discrepancies at checkpoints, averaged across all three algorithms, are reduced by all of them, with the greatest relative reductions occurring for TD3. (Table 1). Mean signed discrepancies across checkpoints and seeds change from +3.127 to +1.926 for DDPG, from −1.121 to −0.384 for TD3, and from −2.082 to −1.085 for SAC. Figure 1 shows why keeping both the signs are useful: the recorded DDPG estimates are greater than the reference, Lower estimated TD3 and SAC. The estimates can vary with number of episodes. At episode 10,000, the mean signed discrepancy changes from 2.573 to 2.422 for DDPG, from −1.041 to −0.078 for TD3, and from −4.043 to −3.631 for SAC. There is still significant residual error in some of the final evaluations and the corrected SAC statistic is highly variable between seeds. These observations are of the recorded diagnostic not an improvement in control performance.

4.1 Findings and Research Vision

The present findings show reduced critic–reference discrepancies and open several opportunities for future research. First, GPR is fitted and evaluated on the same state–action pairs. Future work will examine whether the model can predict signed discrepancies on unseen pairs. Second, Pendulum-v1 uses environment actions in $[-2, 2]$, whereas Stable-Baselines3 critics are trained with normalized actions in $[-1, 1]$ [19]. This mismatch was not corrected, and the results were not recomputed. Future experiments will use consistent action representations. Third, Monte Carlo rollout returns do not always match the critic objectives. Truncated rollouts may omit continuation values, while SAC additionally follows an entropy-regularized objective. Future work will consider rollout references that match these objectives. Fourth, when GPR fitting or prediction fails, the implementation uses $\mu_e = e$ and $\sigma_e = 0$, making the corrected discrepancy zero. Recording these failures separately will help determine their effect on the reported reductions.

These observations define the next research challenges. Reliable residual learning requires independent fitting and evaluation samples, consistent action representations, and rollout references that match the critic objectives. Since GPR predictive variance does not guarantee accurate corrections, uncertainty calibration and out-of-sample validation are also necessary [15]. An important next step is to investigate whether an agent can use these validated residual predictions during critic updates and whether they improve policy learning without introducing instability. Our broader vision is to develop agents that learn not only action values but also the direction, size, and confidence of their own estimation errors. Instead of depending entirely on critic predictions, an agent could learn when its estimated value is too high or too low and determine how much correction is needed. This raises an important question: can an agent learn from its own value-estimation errors and use that knowledge to improve future decisions? Such an approach would move beyond correcting values at evaluation checkpoints toward agents that continually examine and adjust their value estimates during learning.

This idea could have an impact on autonomous systems, where agents must make decisions using learned value estimates. For example, in robotic manipulation, an agent may overestimate the value of an action that leads to a failed grasp or underestimate an action that could complete the task. In autonomous navigation, inaccurate values may cause an agent to prefer an unsuitable route or ignore a better available action. Learning signed discrepancies could help these agents identify both types of errors rather than only reducing overestimated values. The vision becomes more challenging in multiagent systems, where the actions of one agent can change the outcomes experienced by other agents. Consider multiple autonomous robots navigating a shared workspace. Each robot learns its own value estimates, but changes in the movement or decisions of other robots may affect those estimates. An agent may overestimate the value of moving through a crowded path or underestimate the value of waiting for another agent. A future research direction is to investigate whether each agent can learn its own signed critic errors while adapting to the changing behaviour of other agents. Another possibility is to examine whether agents can share information about their estimation errors and uncertainty to support coordination.

Another research direction is to understand how correction parameters affect learning across agents and environments. In Eq. (11), $\tau = 5$ is fixed, while the length-scale ℓ in Eq. (9) controls how GPR models state–action relationships [17, 23]. A correction effective for one critic may not generalize to others. Future research will examine adapting these parameters based on residual errors and predictive uncertainty. This raises the question of when an agent should apply a correction or rely on its original critic estimate. Comparing fixed and adaptive corrections during training could help extend the approach beyond diagnostic experiments.

In multiagent learning, an agent's value estimates also depend on the changing behaviour of other agents. Corrections learned from earlier interactions may become inaccurate as other agents change their policies. A fundamental open challenge is distinguishing critic approximation errors from discrepancies caused by changing policies of other agents, and determining how these sources of error should influence coordination. One direction is to study agent-specific residual models learned from local observations and interaction histories. In cooperative settings, agents could share information about their predicted errors and uncertainty. For example, robots performing a shared task could use this information to update their critics while adapting to other agents' decisions. Future experiments could compare independent and shared residual corrections in terms of policy returns, coordination, and learning stability. The broader question is whether agents can use knowledge of their own estimation errors to improve collective decision-making.

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